



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (21/09/2026 – 27/09/2026)

WEEK 9

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

DESIGN AND IMPLEMENT A CHATBOT THAT CAN ANSWER TO
DIFFERENT TYPE OF QUERY LIKE STRUCTURED, UNSTRUCTURED AND
SEMANTIC KNOWLEDGE BY DECIDING OWN WHAT TYPE OF QUERY BY

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COLLEGE NAME :

BITS PILANI

21/09/2026	- Improved the query-routing logic by providing the classifier with available database schema, document metadata, and tool descriptions instead of relying on fixed query patterns. - Tested automatic routing for structured, unstructured, and multi-source questions.
22/09/2026	- Performed end-to-end testing of the structured SQL pipeline using the uploaded AQL dataset. - Verified Text-to-SQL execution for analytical queries and successfully retrieved the highest PM2.5 value from the dataset.
23/09/2026	- Tested the document RAG pipeline without explicitly specifying the document filename and verified automatic document-source selection. - Evaluated ambiguous and unsupported queries to check clarification handling and hallucination resistance.
24/09/2026	- Tested a multi-source query requiring both SQL computation and document retrieval. - Identified a limitation in cross-tool context transfer where the SQL result was correct, but its computation meaning was not fully preserved during final answer synthesis.
25/09/2026	- Reviewed current project limitations and prepared the next development phase for improving multi-source reasoning, grounding, evaluation, and response latency. - Planned improvements to the existing Streamlit interface, including file upload, organization selection, citations, confidence display, and execution tracing for debugging the agent workflow.
26/09/2026	Holiday (Second Saturday)
27/09/2026	Holiday (Sunday)

Next Week Plan	Improve multi-source reasoning, UI, and debugging support.
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REFERENCE:

https://microsoft.github.io/graphrag/get_started/

https://microsoft.github.io/graphrag/index/default_dataflow/

<https://docs.ollama.com/>

<https://www.sqlite.org/docs.html>

<https://github.com/meta-llama/llama3>

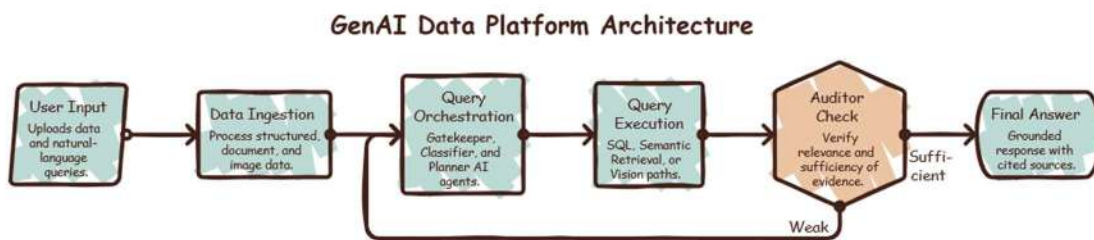
<https://docs.python.org/3/library/sqlite3.html>

<https://arxiv.org/pdf/2201.05880v1>

<https://arxiv.org/pdf/2411.04952>

<https://arxiv.org/pdf/2510.06265v2>

SCREEN SHOTS:



utants in $\mu\text{g}/\text{m}^3$." So, the relationship is that the other pollutants have a unit of $\mu\text{g}/\text{m}^3$. But the text doesn't specify what the unit is for CO. Wait, but they are in $\mu\text{g}/\text{m}^3$. So, the triple would be CO (entity1) has a relationship of "unit" to mg/m^3 , and other pollutants (entity2) has a relationship of "unit" to $\mu\text{g}/\text{m}^3$. But the problem is that the text doesn't explicitly state that other pollutants have $\mu\text{g}/\text{m}^3$ as their unit. It just says "other pollutants in $\mu\text{g}/\text{m}^3$." So, the relationship is that the other pollutants are measured in $\mu\text{g}/\text{m}^3$. But how to represent that in the triple? The entities would be CO, other pollutants, and their units. But the text doesn't mention the other pollutants' units except for CO. Wait, the first part says "CO in mg/m^3 and other pollutants in $\mu\text{g}/\text{m}^3$." So, the "other pollutants" are the ones in $\mu\text{g}/\text{m}^3$. So, the relationship is that CO has a unit of mg/m^3 , and the other pollutants have a unit of $\mu\text{g}/\text{m}^3$. But the text doesn't explicitly state that the other pollutants have that unit. It just says that they are in $\mu\text{g}/\text{m}^3$. So, the triple would be CO (entity1) and its unit (mg/m^3), and the other pollutants (entity2) and their unit ($\mu\text{g}/\text{m}^3$). But the problem is that the text doesn't mention the other pollutants' units except for CO. Wait, the text says "other pollutants in $\mu\text{g}/\text{m}^3$." So, the "other pollutants" are the ones with the unit $\mu\text{g}/\text{m}^3$. So, the relationship is that the other pollutants have a unit of $\mu\text{g}/\text{m}^3$. But how to represent that in the triple? The entities would be the pollutant (other pollutants) and their unit. But the text doesn't list the other pollutants, just mentions that they are in $\mu\text{g}/\text{m}^3$. So, the triple would be: CO (entity1) has a unit of mg/m^3 , and other pollutants (entity2) have a unit of $\mu\text{g}/\text{m}^3$. But the confidence would be 1 because the text explicitly states that CO is in mg/m^3 and other pollutants are in $\mu\text{g}/\text{m}^3$. Wait, but the text says "CO in mg/m^3 and other pollutants in $\mu\text{g}/\text{m}^3$." So, the "other pollutants" are the ones in $\mu\text{g}/\text{m}^3$. So, the relationship between CO and its unit is explicit, and the relationship between other pollutants and their unit is also explicit. So, two triples: one for CO and its unit, and another for other pollutants and their unit. But the problem is that the text doesn't mention the other pollutants' unit except for CO. Wait, the text says "other pollutants in $\mu\text{g}/\text{m}^3$." So, the "other pollutants" are the ones with the unit $\mu\text{g}/\text{m}^3$. So, the triple would be: CO ('

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you > /ingest C:\Users\ADITYA SINGH\Downloads\BISAG-CHATBOT\test\struct.csv
Ingested struct.csv -- 1 table(s) written [struct] in 0.1s

you > Which city has the highest average PM2.5 value in the uploaded dataset refer to struct.csv here?
Thinking... Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-09-22 15:44:40 | WARNING | huggingface_hub.utils._http | Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
2026-09-22 15:44:42 | WARNING | app.models.llm | LLM load attempt failed (4bit): Using 'bitsandbytes' 4-bit quantization requires bitsandbytes:
'pip install -U bitsandbytes>=0.46.1'
Loading weights: 100% [REDACTED] 311/311 [00:03<00:00, 101.48it/s]
2026-09-22 15:44:48 | WARNING | app.models.llm | Could not load Qwen/Qwen3-1.7B in 4bit (4bit: ImportError: Using 'bitsandbytes' 4-bit quantization requires bitsandbytes: 'pip install -U bitsandbytes>=0.46.1'). Loaded unquantized instead, which uses more memory.
Datahall
The city with the highest average PM2.5 value in the dataset is Gurugram,
with an average of 418.0.

[structured] [confidence 5/5] [via sql_analyst_tool] [23.7s]
Sources:
- [Source: struct]

you >

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